Hands on Simple Linear Regression

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COMP6577 - Machine Learning >>> Bina Nusantara University

Importing the libraries

In [1]:
```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
```

Importing the dataset

In [2]:
```
dataset = pd.read_csv("https://raw.githubusercontent.com/joh4nnes/MLData/main/Salary_Data.csv")
dataset.head()
```

Out[2]:

	YearsExperience	Salary
0	1.1	39343.0
1	1.3	46205.0
2	1.5	37731.0
3	2.0	43525.0
4	2.2	39891.0

In [3]:
```
# 1. Find the dimensions of the dataset
dataset.shape
```

Out[3]:
```
(30, 2)
```

In [4]:
```
# 2. Look at the data types of the features
dataset.dtypes
```

Out[4]:
```
YearsExperience    float64
Salary             float64
dtype: object
```

In [5]:
```
# 3. Look at statistical summary of the features
dataset.describe().T
```

Out[5]:

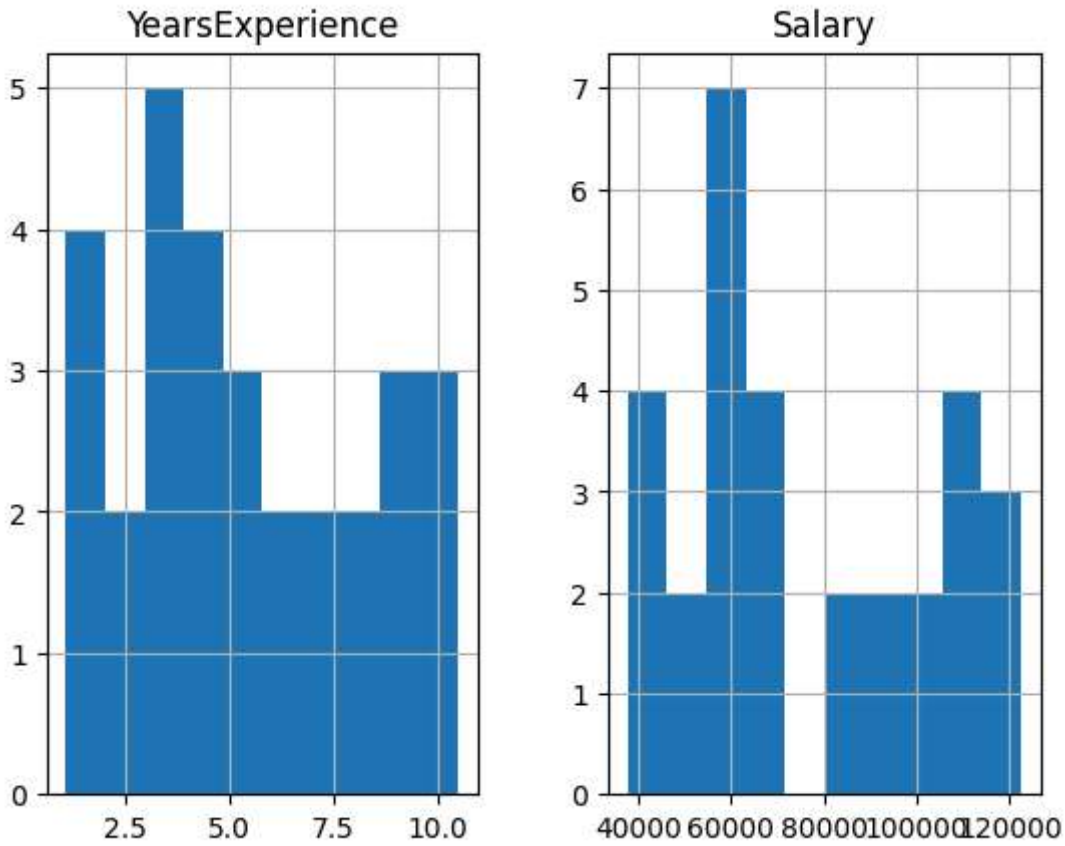
	count	mean	std	min	25%	50%	75%	max
YearsExperience	30.0	5.313333	2.837888	1.1	3.20	4.7	7.70	10.5
Salary	30.0	76003.000000	27414.429785	37731.0	56720.75	65237.0	100544.75	122391.0

In [6]:
```
# 4. Look for missing values in the features
np.round(dataset.isnull().mean() * 100,1)
```

Out[6]:
```
YearsExperience    0.0
Salary             0.0
dtype: float64
```

In [7]:
```
# 5. Looking at distributions of numeric features
# create histogram for numeric data
dataset.hist()

# show plot
plt.show()
```



In [8]:
```
#6. Independent and Dependent Variable
X = dataset.iloc[:, :-1].values
y = dataset.iloc[:, -1].values
```

Splitting the dataset into the Training set and Test set

In [9]:
```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 1/3, random_state = 0)
```

Training the Simple Linear Regression model on the Training set

In [10]:
```
from sklearn.linear_model import LinearRegression
regressor = LinearRegression()
regressor.fit(X_train, y_train)
```

Out[10]:
```
▾ LinearRegression
LinearRegression()
```

Predicting the Test set results

In [11]:
```
y_pred = regressor.predict(X_test)
```

Visualising the Training set results

In [12]:
```
plt.scatter(X_train, y_train, color = 'red')
plt.plot(X_train, regressor.predict(X_train), color = 'blue')
plt.title('Salary vs Experience (Training set)')
plt.xlabel('Years of Experience')
plt.ylabel('Salary')
plt.show()
```

Visualising the Test set results

In [13]:
```
plt.scatter(X_test, y_test, color = 'red')
plt.plot(X_train, regressor.predict(X_train), color = 'blue')
plt.title('Salary vs Experience (Test set)')
plt.xlabel('Years of Experience')
plt.ylabel('Salary')
plt.show()
```

source : https://www.javatpoint.com/simple-linear-regression-in-machine-learning